COMP1661 Logbook Upload Template

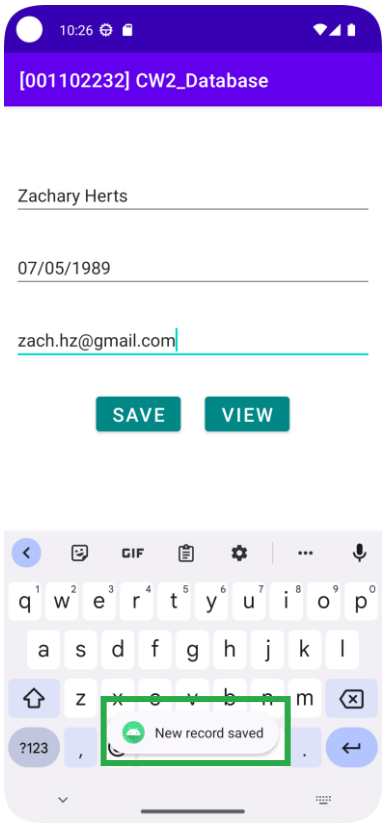
1. Basic Information

1.1 Student name	**Marzuq Sogbanmu**
1.2 Who did you work with? Note that for logbook exercises you are allowed to work with one other person as long as you give their name and login id and both contribute to the work.	**Name:** **Login id:**
1.3 Which Exercise is this? Tick as appropriate.	• Exercise 1 ☐ • Exercise 2 ☐ • Exercise 3 ☒
1.4 How well did you complete the exercise? Tick as appropriate.	• I tried but couldn't complete it ☐ • I did it but I feel I should have done better ☒ • I did everything that was asked ☐ • I did more than was asked for ☐
1.5 Briefly explain your answer to question 1.4	**I was able to achieve functionality for entering user details, saving them to the database, and view saved records. However, coding the functionality for the avatars was proving to be much more difficult than anticipated, therefore I was not able to implement the feature in time.**

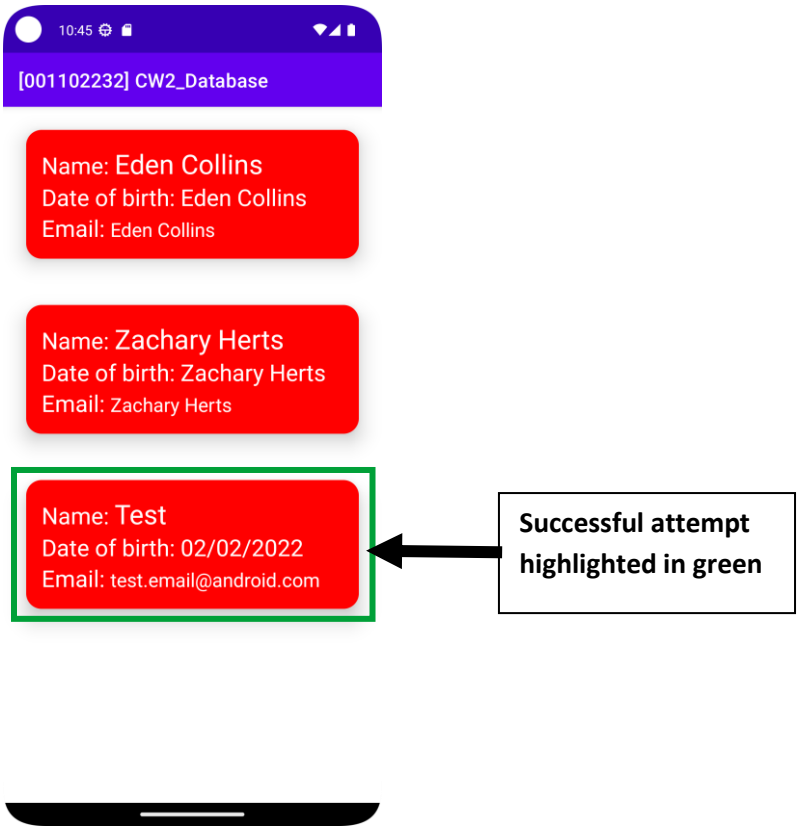
2. Exercise answer

2.1 Screen shots demonstrating what you achieved

Saving a record



Viewing of records



2.2 Code that you wrote

Copy and paste relevant code here. Actual code please, not screen shots.

You need to add brief explanation.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:padding="10dp"
    >


    <EditText
        android:id="@+id/et_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="60dp"
        android:ems="10"
        android:hint="Name"
        android:inputType="textMultiLine" />

    <EditText
        android:id="@+id/et_dob"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/et_name"
        android:layout_marginTop="30dp"
        android:ems="10"
        android:hint="Date of birth (DD/MM/YYYY)"
        android:inputType="textMultiLine" />

    <EditText
        android:id="@+id/et_email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/et_dob"
        android:layout_marginTop="30dp"
        android:ems="10"
        android:hint="Email"
        android:inputType="textMultiLine" />


    <LinearLayout
        android:id="@+id/ll_buttons"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/et_email"
        android:layout_marginTop="30dp"
        android:gravity="center">

        <Button
            android:id="@+id/btn_save"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/et_email"
            android:backgroundTint="@color/teal_700"
            android:layout_marginEnd="25dp"
            android:text="Save"
            android:textAlignment="center"
            android:textSize="20sp" />

        <Button
            android:id="@+id/btn_view"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/et_email"
            android:backgroundTint="@color/teal_700"
            android:text="View"
            android:textAlignment="center"
            android:textSize="20sp" />
    </LinearLayout>

</RelativeLayout>
```

Activity_users.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Users">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/rv_users"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Userrecords.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    app:cardElevation="12dp"
    app:cardCornerRadius="16dp"
    android:layout_margin="24dp"
    android:backgroundTint="#FF0000">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:orientation="vertical">

<!--        Linear layout for name-->
        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:layout_gravity="center_vertical">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Name: "
                android:textSize="24dp"
                android:textColor="@color/white"
                style="bold"/>
            <TextView
                android:id="@+id/tv_record_name"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
```

```
                android:textSize="28dp"
                android:textColor="@color/white"
                style="bold"/>
        </LinearLayout>

<!--        Linear layout for DOB-->
        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:layout_gravity="center_vertical">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Date of birth: "
                android:textSize="24dp"
                android:textColor="@color/white"
                style="bold"/>
            <TextView
                android:id="@+id/tv_record_dob"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:textSize="24dp"
                android:textColor="@color/white"
                style="bold"/>
        </LinearLayout>

<!--        Linear layout for email-->
        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:layout_gravity="center_vertical">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Email: "
                android:textSize="24dp"
                android:textColor="@color/white"
                style="bold"/>
            <TextView
                android:id="@+id/tv_record_email"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:textSize="20dp"
                android:textColor="@color/white"
                style="bold"/>
        </LinearLayout>


    </LinearLayout>

</androidx.cardview.widget.CardView>
```

MainAcitvity.java

```
package com.example.a001102232cw2_database;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
```

```
import android.widget.EditText;
import android.widget.Toast;

import java.io.BufferedReader;
import java.io.FileInputStream;
import java.io.FileNotFoundException;
import java.io.FileOutputStream;
import java.io.InputStreamReader;

public class MainActivity extends AppCompatActivity {

    EditText name, dob, email;
    Button save, view;
    DBHelper db;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        name = findViewById(R.id.et_name);
        email = findViewById(R.id.et_email);
        dob = findViewById(R.id.et_dob);
        save = findViewById(R.id.btn_save);
        view = findViewById(R.id.btn_view);

        db = new DBHelper(this);

//      Starting a new activity by clicking the View button
        view.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                startActivity(new Intent(MainActivity.this, Users.class));
            }
        });

//        Functionality for saving new records
        save.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String name_text = name.getText().toString();
                String dob_text = dob.getText().toString();
                String email_text = email.getText().toString();

                Boolean check_saved_data = db.insert_data(name_text, dob_text,
email_text);
                if (check_saved_data==true) {
                    Toast.makeText(MainActivity.this, "New record saved",
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Record failed to save",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
```

DBHelper.java

```
package com.example.a001102232cw2_database;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;
```

```
import androidx.annotation.Nullable;

public class DBHelper extends SQLiteOpenHelper {
    private static final String DB_name = "details";
    public static final String clm_ID = "user_id";
    public static final String clm_Name = "name";
    public static final String clm_DOB = "dob";
    public static final String clm_Email = "email";

    private  SQLiteDatabase database;

    private static final String DATABASE_CREATE = String.format(
      "CREATE TABLE %s (" +
      " %s INTEGER PRIMARY KEY AUTOINCREMENT, " +
      " %s TEXT, " +
      " %s TEXT, " +
      " %s TEXT)",
      DB_name, clm_ID, clm_Name, clm_DOB, clm_Email);

    public DBHelper(Context context) {
        super(context, DB_name, null, 1);
        database = getWritableDatabase();
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(DATABASE_CREATE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + DB_name);

        Log.w(this.getClass().getName(), DB_name + " database upgrade to version, " +
newVersion + "old data lost");
        onCreate(db);
    }

    public Boolean insert_data (String name, String dob, String email) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(clm_Name, name);
        contentValues.put(clm_DOB, dob);
        contentValues.put(clm_Email, email);

        long result = db.insert(DB_name, null, contentValues);

        if (result==-1){
            return false;
        }
        else {
            return true;
        }
    }

    public Cursor get_Data () {
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cursor = db.rawQuery("SELECT name, dob, email FROM details", null);
        return cursor;
    }
}
```

Adapter.java

```
package com.example.a001102232cw2_database;

import android.content.Context;
```

```
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.ArrayList;

public class Adapter extends RecyclerView.Adapter<Adapter.ViewHolder> {
    private Context context;
    private ArrayList name_id, dob_id, email_id;

    public Adapter(Context context, ArrayList name_id, ArrayList dob_id, ArrayList
email_id) {
        this.context = context;
        this.name_id = name_id;
        this.dob_id = dob_id;
        this.email_id = email_id;
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        View v =
LayoutInflater.from(context).inflate(R.layout.userrecord,parent,false);
        return new ViewHolder(v);
    }

    @Override
    public void onBindViewHolder(@NonNull ViewHolder holder, int position) {
        holder.name_id.setText(String.valueOf(name_id.get(position)));
        holder.dob_id.setText(String.valueOf(dob_id.get(position)));
        holder.email_id.setText(String.valueOf(email_id.get(position)));
    }

    @Override
    public int getItemCount() {
        return name_id.size();
    }

    public class ViewHolder extends RecyclerView.ViewHolder {
        TextView name_id, dob_id, email_id;
        public ViewHolder(@NonNull View itemView) {
            super(itemView);
            name_id = itemView.findViewById(R.id.tv_record_name);
            dob_id = itemView.findViewById(R.id.tv_record_dob);
            email_id = itemView.findViewById(R.id.tv_record_email);
        }
    }
}
```

Users.java

```
package com.example.a001102232cw2_database;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.Toast;

import java.util.ArrayList;
```

```java
public class Users extends AppCompatActivity {
    RecyclerView recyclerView;
    ArrayList<String> name, dob, email;
    DBHelper db;
    Adapter adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_users);
        db = new DBHelper(this);
        name = new ArrayList<>();
        dob = new ArrayList<>();
        email = new ArrayList<>();
        recyclerView = findViewById(R.id.rv_users);
        adapter = new Adapter(this, name, dob, email);
        recyclerView.setAdapter(adapter);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));
        display_data();
    }

    private void display_data() {
        Cursor cursor = db.get_Data();
        if (cursor.getCount()==0) {
            Toast.makeText(Users.this, "No record exists", Toast.LENGTH_SHORT).show();
            return;
        } else {
            while (cursor.moveToNext()) {
                name.add(cursor.getString(0));
                dob.add(cursor.getString( 1));
                email.add(cursor.getString( 2));
            }
        }
    }
}
```